

IISER Mohali -PHY102-Electromagnetism

Assignment 2

Due date: 6th Feb 2025

Q1. Find the electric field and potential inside a conducting charged sphere and outside it . draw the nature of field and potential.

Q2. Consider two infinite, parallel conducting plates, spaced a distance d apart. The region between the plates is filled with a dielectric ϵ .The capacitor is connected with 5 volt. Find the expression for potential (V) and the electric field within the plates (E) where 5 volt plate is at $z=-d$ and ground plate is at $z=0$.

Q3. A charge of "q" is placed at $z=d$ distance above the 2D (x-y plane) infinite conducting metal plate, and it is grounded. Derived the expression for Electric Field (E), surface charge density on metal plate and potential (V) at point (x,y,z) above the plate. what will be the electric field below the plate?.

Q4. A point charge is placed outside a grounded conducting sphere. Find the potential at the surface:Find the actual charge density induced on the surface of grounded sphere.

Q5. A fixed ring of radius "r" has a positive charge "Q" uniformly distributed over it. A particle of negative charge e and mass m is placed on the axis of the ring at a small distance from the centre of the ring. Show that the particle, if free to move, will undergo a simple harmonic motion. Find the time period of oscillation.

Q6. The electric field in a certain region is given as $\vec{E} = A r^3 \hat{r}$. Prove that charge contained within a spherical surface of radius 'a' centred at the origin is $4\pi\epsilon A a^5$